

Bounds on Multicolor Ramsey Numbers

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NYC Discrete Math REU

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Party Problem

Question

What is the minimum number of people needed at a party to guarantee that either three people know each other, or three people do not?

Party Problem

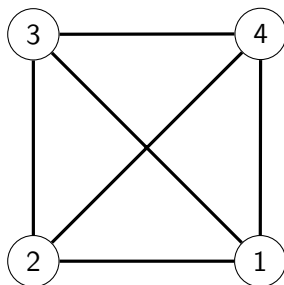
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What is the minimum number of people needed at a party to guarantee that either three people know each other, or three people do not?

Whatever number this ends up being will forever be known as $r(3)$.

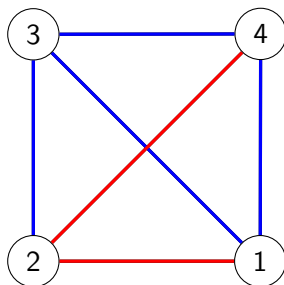
Exploring this question with four people

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- We can define a relationship between these people by coloring the edges. Thinking back to the question, we were interested in three friends or three strangers. So, we can 2-color this graph using blue and red to represent friends and strangers, respectively.

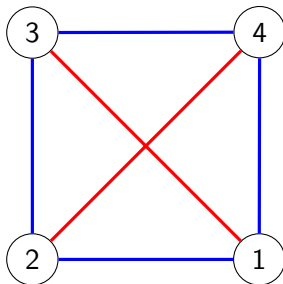


What should we be looking for?

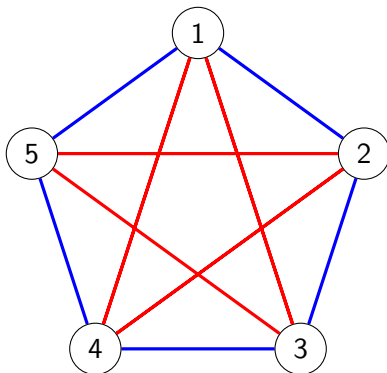
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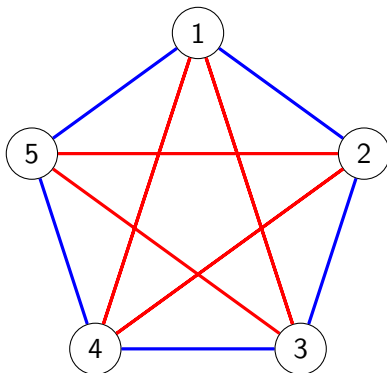
- All that is needed is to find one of these graphs where we do not have a monochromatic triangle.
- Here is such a graph on four vertices.



What about five people?



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Not enough either.

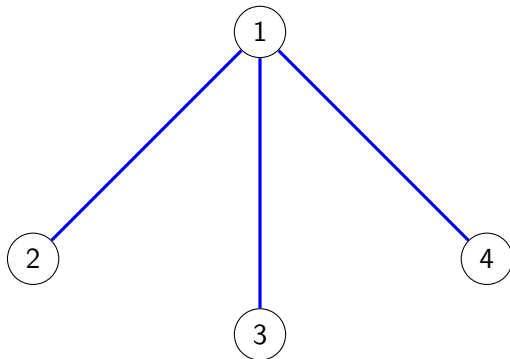
Six people will suffice

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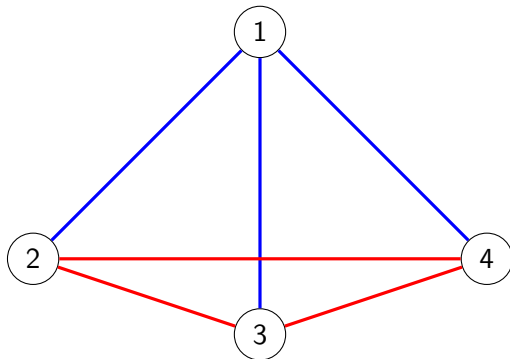
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For example, $r(3; 3) = 17$ is a known multicolor Ramsey number.

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These bounds (that came from a mixture of arguments from Paul Erdős, George Szekeres and Hanno Lefmann) were not improved upon for a very long time.

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- Finally, the best known bound came a few months ago from Will Sawin,

$$r(t; \ell) \geq e^{0.266027(\ell-2)t} 2^{\frac{t}{2}+o(t)} \approx 1.305^{t\ell}.$$

Plans for this summer

Bipartite Ramsey Number.

The bipartite Ramsey number $b(t; \ell)$ is the least positive integer N such that every ℓ -valued $N \times N$ matrix contains a $t \times t$ submatrix with equal entries.

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Our goal is to use techniques that led to these recent improvements on the lower bound of $r(t; \ell)$ to find a new lower bound for $b(t; \ell)$.

References

- Yuval Wigderson. Lecture notes on multicolor Ramsey numbers.
[http://web.stanford.edu/](http://web.stanford.edu/yuvalwig/math/teaching/MulticolorRamsey.pdf)
[yuvalwig/math/teaching/MulticolorRamsey.pdf](http://web.stanford.edu/yuvalwig/math/teaching/MulticolorRamsey.pdf),
- Yuval Wigderson. An improved lower bound on multicolor Ramsey numbers. arXiv:2009.12020, 2020.